SAHIL PATEL

Backend Engineer · Go · Node.js

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Education

Punjab Technical University — Jalandhar, Punjab
B.Tech in Computer Science Engineering — 2022 – 2026 (Completed June 2026)

Skills

Languages: Go, JavaScript (Node.js)
Backend: Gin, GORM, Express.js, REST APIs, gRPC, Protobuf, WebSockets, JWT, RBAC, Middleware, Microservices
Databases: PostgreSQL, MongoDB, MySQL
DevOps: Docker, Google Cloud Platform (GCP), GitHub
Tools: Git/GitHub, Postman, Linux

Projects

Concurrent Database Engine [GitHub] *Go*

- Built a lightweight JSON database engine from scratch with concurrent CRUD handling **10,000+ ops/sec** using goroutines, channels, and mutex synchronisation — zero data races under load; atomic writes via os.Rename ensure crash-safe persistence.
- Demonstrates deep practical understanding of **Go concurrency** primitives (goroutines, channels, sync.Mutex) applied to a real systems design problem.

Distributed Chat Platform [GitHub] *Go — Gin — PostgreSQL — MongoDB — WebSockets — JWT*

- Designed and implemented a production-ready real-time chat backend supporting **500+ concurrent WebSocket connections** across one-to-one and group conversations, with live presence tracking and message delivery guarantees.
- Architected a hybrid storage layer: **PostgreSQL** for relational data (users, rooms, participants) and **MongoDB** for high-volume message persistence, sustaining **1,000+ messages/sec** write throughput.
- Implemented **JWT** auth, RBAC, middleware-based authorisation, and pagination-optimised message retrieval for large conversation histories.
- Designed a clean Handler–Service–Repository (HSR) architecture enabling modular microservice extraction without rewrites; added context propagation and graceful shutdown handling.

Experience

Smollan India (Google Retail Hub) — Remote
Golang Backend Developer Intern — Apr 2026 – Jun 2026

- Developed and shipped **15+** production REST and gRPC APIs powering the Google Retail Hub platform using Go, Gin, and GORM, supporting daily operations across **10+** business modules.
- Reduced API response time from **480ms to 385ms (20% improvement)** by identifying ORM bottlenecks and rewriting critical paths with optimised raw SQL and PostgreSQL indexes.
- Implemented JWT authentication, RBAC, and middleware-driven authorisation securing endpoints across user, inventory, and reporting modules.
- Wrote unit and integration tests for critical API paths; built a Postman collection of **100+** test cases and resolved **25+** production bugs within Agile sprint cycles.

Docimize Tech Solutions — Remote
Backend Developer Intern — Jul 2025 – Sep 2025

- Built scalable REST APIs using **Node.js**, Express.js, and MongoDB serving **1,000+** daily requests with sub-200ms p95 response time.
- Reduced API latency from **400ms to 200ms (50% improvement)** by eliminating N+1 query patterns, adding compound indexes, and batching redundant database calls.
- Optimised MongoDB schemas and indexing strategy, cutting average query execution time by **30%** across high-frequency read endpoints.